DELHI TECHNOLOGICAL UNIVERSITY

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CRISISFLOW

Multi-Agent AI for Autonomous Urban Disaster Response and Emergency Mobility Optimization

DETAILED RESEARCH PLAN

B.Tech. Computer Science & Engineering — Major Project

PROJECT IDENTIFIER	095
PROJECT GUIDE	Ms. Anjali Bansal
PROJECT TEAM	Shubham Singh Negi (23/CS/407) Shivansh Dharmshaktu (23/CS/399) Samar Pratap Singh (23/CS/369)
JUNIOR TEAM MEMBER	Hardik Gupta (25/A02/063)
CO-GUIDE	Not specified
SUBMISSION CONTEXT	Final-Year UG Research Plan

Prepared with reference to the DTU CSE Detailed Research Plan submission notice dated 29 September 2026 and the supplied CrisisFlow project concept document.

DOCUMENT STRUCTURE

This plan is organized to address the department's requested submission components and to keep the proposed work experimentally testable.

1. Project Overview and Abstract
2. Problem Statement, Motivation and Objectives
3. Scope, Assumptions and Research Questions
4. Brief Literature Review
5. Research Gap and Proposed Contribution
6. Proposed System Architecture and Methodology
7. Experimental Design and Evaluation
8. Technical Requirements
9. Expected Outcomes, Risks and Mitigation
10. Work Plan and Gantt Chart
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Scope note. The primary research track is emergency-aware multi-agent traffic-signal coordination and dynamic routing in simulation. Computer vision, learned traffic forecasting and a GenAI operator interface are treated as extensions, not prerequisites for the core experimental claim.

1. PROJECT OVERVIEW AND ABSTRACT

Project title: CrisisFlow — Multi-Agent AI for Autonomous Urban Disaster Response and Emergency Mobility Optimization
Project identifier: 095
Guide: Ms. Anjali Bansal

Abstract. Urban emergencies such as road accidents, fires, flooding and infrastructure disruption create a time-critical mobility problem. Conventional shortest-path navigation and fixed or locally adaptive traffic signals may fail to coordinate multiple emergency vehicles when road availability, congestion and competing incidents change during response. Emergency vehicles can also impose secondary delays on ordinary traffic if signal priority and rerouting are handled independently. CrisisFlow proposes a simulation-based, emergency-aware decision framework that jointly studies dynamic emergency routing and coordinated traffic-signal control using multi-agent reinforcement learning (MARL). Intersections are modelled as agents that observe local queues, signal phase, nearby emergency-vehicle status, estimated arrival time and relevant neighbouring congestion. A routing component updates emergency paths when travel costs or road availability change. The system will be evaluated in Simulation of Urban MObility (SUMO), using reproducible scenarios with varying demand, road blockages and single or simultaneous emergencies. Performance will be compared against fixed-time and actuated signal control, shortest-path routing, and suitable single-agent or independent-agent reinforcement-learning baselines. The principal measures will include emergency travel/response time, ordinary-vehicle delay, queue length, throughput, rerouting stability and incident success rate. The intended contribution is an experimentally evaluated coordination strategy and reward design that balances emergency priority with network-wide disruption, rather than assuming that improvements occur in advance.

Keywords: Multi-agent reinforcement learning; emergency vehicle routing; traffic-signal control; disaster response; SUMO; dynamic routing; congestion management.

Brief description

CrisisFlow is envisioned as an AI-assisted decision system for a city's emergency mobility network. It monitors a simulated road network, reacts to emergency events and changing road conditions, coordinates signals along and around emergency routes, and reroutes ordinary fleet traffic only when the expected benefit justifies the disruption. The initial research prototype will prioritize a robust simulation and a fair baseline comparison before optional perception or operator-facing components are added.

2. PROBLEM STATEMENT, MOTIVATION AND OBJECTIVES

2.1 Problem statement

How can multiple autonomous agents coordinate emergency-vehicle routes and urban traffic signals under dynamic road and disaster conditions so that emergency travel time is reduced while secondary congestion and disruption to ordinary traffic remain limited?

2.2 Motivation

A route that is fastest at dispatch time may become suboptimal after a road closure, an unexpected queue or another emergency. Similarly, granting a green phase to one emergency vehicle can increase queues at neighbouring approaches. If routing and signal pre-emption are optimized separately, each component may make locally reasonable decisions that produce poor network-wide outcomes. CrisisFlow therefore treats the road network as a coupled, time-varying system and evaluates the trade-off between urgent response and normal mobility.

2.3 Objectives

- Build a repeatable urban traffic simulation in SUMO with ordinary traffic, emergency vehicles, signalized intersections and configurable incidents.
- Implement a dynamic emergency-routing baseline that accounts for current edge travel time and road closures.

- Develop a MARL-based traffic-signal controller whose state includes traffic queues, current phase and emergency proximity/ETA; investigate neighbour information for coordination.
- Define an emergency-aware multi-objective reward that penalizes emergency delay, ordinary traffic delay, excessive queues and unstable route/signal changes.
- Evaluate the proposed controller against transparent baselines across demand levels and emergency scenarios using repeated simulation seeds.
- Report limitations, ablation results and failure cases, and provide a demonstration that visualizes route and signal decisions.

3. SCOPE, ASSUMPTIONS AND RESEARCH QUESTIONS

3.1 Core scope

- Simulation-first prototype; no real-world traffic-signal actuation or safety-critical deployment is proposed.
- Initial network: one manageable urban grid or a selected SUMO/OSM-derived network, expanded only after the basic pipeline is stable.
- Emergency classes: ambulance and fire engine first; police/rescue vehicles may be represented if scenario complexity permits.
- Incidents: road closure/blockage and multiple emergency requests; flood/fire hazard zones can be represented as unavailable or high-cost edges.
- Primary control: intersection-level signal decisions plus dynamic route updates. Fleet rerouting is a secondary module if time permits.

3.2 Out of scope for the minimum viable research

Training a production-grade vision model, deploying roadside sensors, live integration with municipal systems, physical signal control and a full conversational GenAI command centre are not required to validate the core research question. Camera-based vehicle detection (e.g., YOLO), learned traffic forecasting (e.g., XGBoost/LSTM/Transformer) and a GenAI explanation interface can be added only after the simulation and baseline evaluation are reliable.

3.3 Research questions

- RQ1: Does coordinated emergency-aware MARL reduce emergency travel time compared with fixed-time/actuated signals and conventional shortest-path routing?
- RQ2: What is the effect on ordinary-vehicle delay, queue length and network throughput when emergency priority is enabled?
- RQ3: Does sharing local information between neighbouring intersection agents improve performance over independent agents?
- RQ4: How robust is the controller when the number of emergencies, traffic demand or road closures differ from training scenarios?
- RQ5: Which reward components contribute most to the emergency-time versus network-disruption trade-off?

4. BRIEF LITERATURE REVIEW

The selected works are recent and directly related to reinforcement learning, emergency routing, traffic-signal coordination or disaster evacuation. The literature suggests that routing and signal control should be treated as coupled decisions, while network-wide traffic effects and generalization to unfamiliar emergencies require explicit evaluation.

4.1 Emergency routing and signal pre-emption with MARL

Su et al.'s EMVLight framework jointly considers dynamic emergency-vehicle routing and traffic-signal pre-emption in a decentralized reinforcement-learning setting. The work is closely aligned with CrisisFlow because it treats route choice and signal control as coupled decisions and reports both emergency and non-emergency travel-time outcomes. It motivates a baseline that measures network-wide effects rather than optimizing emergency passage alone. CrisisFlow's intended extension is to test the same general trade-off under explicit disaster events, simultaneous emergency requests and road closures, while keeping the implementation scope suitable for a B.Tech project.

4.2 Reinforcement-learning-based routing on large street networks

Thompson, Gao and Behzadan present a reinforcement-learning routing approach for large street networks in the context of evacuation and emergency routing. Its emphasis on changing traffic and hazardous/safe-route criteria is relevant to representing blocked roads and disaster-affected edges. CrisisFlow can use this line of work to inform its routing component and scenario design. The project will not assume that a learned route is automatically safer or faster; the route policy will be evaluated against shortest-path and time-dependent routing baselines under the same simulated demand.

4.3 Distributed cooperation for network-wide traffic-signal control

SocialLight studies distributed cooperation among traffic-signal agents, using local neighbourhood information to estimate agents' contributions and improve scalability. Although it is not specifically an emergency-routing paper, it informs the multi-intersection control layer of CrisisFlow. It suggests that purely isolated controllers may fail to coordinate their impact on adjacent junctions. CrisisFlow will therefore compare independent agents with a coordinated variant where feasible, and will track ordinary traffic delay as well as emergency performance.

4.4 Congestion-aware multi-agent wildfire evacuation routing

Raei, Safarzadeh and Wang propose congestion-aware MARL for wildfire evacuation routing. Their work is directly relevant to disaster routing because it considers route dispersion, road capacity and hazard avoidance rather than only selecting the individually shortest path. It motivates including road/zone restrictions and congestion penalties in CrisisFlow's scenario model. The project differs in emphasis by focusing on emergency responder mobility and signal coordination alongside ordinary traffic, rather than mass evacuation alone.

4.5 Meta-reinforcement learning for rapid post-emergency adaptation

Jiang et al. combine contrastive predictive coding with multi-agent meta-reinforcement learning for adaptive traffic-signal control after emergencies. This recent work highlights the challenge of responding to unfamiliar traffic patterns and the potential value of learned state representations and policy adaptation. For the initial CrisisFlow prototype, meta-learning and contrastive representation learning are considered advanced extensions. The first experimental priority is a stable MARL baseline, scenario variation and a reproducible comparison; only then should adaptation methods be considered.

4.6 Synthesis

Research direction	Useful insight for CrisisFlow	Implication for proposed work
Emergency routing + signal control	The two decisions influence each other.	Evaluate routing and signal control jointly, not as isolated features.
Large-network RL routing	Routes need to account for dynamic costs and hazards.	Model road closures and compare with shortest-path routing.
Distributed traffic-signal MARL	Neighbour cooperation can improve network-level behaviour.	Compare independent and coordinated agents.
Wildfire evacuation MARL	Congestion-aware route distribution matters during disasters.	Include hazard avoidance and edge-capacity effects.
Meta-RL after emergencies	Unseen scenarios and rapid adaptation remain important.	Test generalization; treat meta-learning as an extension.

The five works are complementary rather than interchangeable: EMVLight informs emergency routing/signal coupling; ReinforceRouting and the wildfire study inform disaster-aware route choice; SocialLight informs cooperative signal agents; and the meta-RL study informs future adaptation to unseen emergency conditions.

5. RESEARCH GAP AND PROPOSED CONTRIBUTION

5.1 Research gap to be investigated

Existing work provides strong components for emergency vehicle routing, signal pre-emption, cooperative traffic control and disaster evacuation. However, the supplied project concept identifies a system-level question: how to coordinate heterogeneous emergency missions when road infrastructure is disrupted and multiple vehicles compete for limited network capacity, while avoiding excessive delays for ordinary traffic. CrisisFlow will investigate this question in a controlled simulation and will explicitly report both sides of the trade-off. This is a project hypothesis to test, not a claim that the proposed system is already superior to existing methods.

5.2 Proposed contribution

- An integrated SUMO testbed for emergency-aware routing and traffic-signal control under changing traffic and road availability.
- A reproducible comparison of conventional baselines, independent RL agents and a coordinated emergency-aware MARL policy, subject to implementation feasibility.
- A multi-objective reward formulation that makes the trade-off between emergency mobility and ordinary traffic disruption explicit.
- Scenario-based robustness analysis covering different traffic demand levels, road blockages and single versus simultaneous emergencies.
- An interpretable demonstration showing the simulated emergency, chosen route, signal decisions and measured network effects.

5.3 Proposed reward design

The controller will be trained with a weighted reward/penalty structure. A candidate formulation is:

$$R_t = -\alpha \cdot D_{emergency} - \beta \cdot D_{ordinary} - \gamma \cdot Q - \delta \cdot C_{reroute} - \eta \cdot V$$

where $D_{emergency}$ denotes emergency-vehicle delay or travel-time cost, $D_{ordinary}$ is ordinary-vehicle delay, Q is queue/congestion cost, $C_{reroute}$ penalizes unnecessary route changes, and V penalizes unsafe or invalid signal transitions if modelled. The weights α, β, γ, δ and η will be tuned on validation scenarios and reported. The reward is a starting design, not a final empirically validated equation; ablation tests will be used to assess the effect of selected terms.

5.4 Hypotheses

- H1: Coordinated emergency-aware control will reduce mean emergency travel time relative to fixed-time signals plus shortest-path routing in at least some tested scenarios.
- H2: A reward that includes ordinary-vehicle delay will produce a better emergency/ordinary mobility trade-off than an emergency-only reward.
- H3: Benefits will vary by network and demand level; performance may degrade under unfamiliar or severe blockages, which will be reported rather than hidden.

6. PROPOSED SYSTEM ARCHITECTURE AND METHODOLOGY

6.1 Architecture

Layer	Inputs	Function / outputs
1. Scenario and network	Road graph, demand, signal layout, incident schedule	Creates repeatable SUMO scenario; supports closures and emergency dispatch.
2. State extraction	TraCI observations: queues, edge travel times, phases, vehicle locations	Builds per-intersection and per-emergency state vectors.
3. Dynamic routing	Current travel costs, blocked edges, destination, emergency priority	Computes or updates emergency routes; invalidates blocked segments.
4. MARL signal control	Local queues, phase, emergency ETA, neighbour summary	Chooses phase/green-extension actions subject to legal signal constraints.
5. Simulation feedback	Actions applied to SUMO	Returns updated traffic state; logs each episode and incident.
6. Evaluation and visualization	Trip, queue, delay, route and event logs	Computes metrics, plots results and presents a scenario replay.

Conceptual flow: Scenario/incident generation → SUMO state observation → dynamic routing + intersection-agent decisions → actions through TraCI → updated simulation state → metric logging and comparison.

6.2 Method

- **Step 1 — Build the environment:** Select a small, reproducible road network; validate routes and signal phases; configure ordinary traffic and emergency vehicle types.
- **Step 2 — Add incident logic:** Schedule an accident/fire/flood-style event, close or penalize selected edges, and dispatch one or more emergency vehicles.
- **Step 3 — Implement reference policies:** Fixed-time signals, actuated control if available, shortest-path emergency routing and a non-learning emergency-priority heuristic.
- **Step 4 — Implement MARL:** Represent each signalized intersection as an agent. Begin with a manageable algorithm (e.g., PPO/MAPPO or independent PPO/DQN depending on action/state design) and a limited discrete action space.
- **Step 5 — Train and validate:** Use separate training, validation and test scenario seeds. Tune reward weights only on training/validation runs.
- **Step 6 — Evaluate:** Run all policies on identical test scenarios and seeds; aggregate results with means and variability; report failure cases and ablations.
- **Step 7 — Demonstrate:** Provide a replay/dashboard showing incident onset, route changes, signal actions and key metrics.

6.3 State, action and reward design

Component	Initial design
Agent	One traffic-signal intersection; optional regional/neighbour communication.
Observation/state	Per-approach queue length or occupancy, current phase, elapsed phase time, nearby emergency present, estimated emergency arrival time, neighbouring congestion summary.
Action	Choose among valid signal phases or a constrained action such as maintain/extend/switch. Enforce minimum green and safe phase-transition rules.
Reward	Weighted penalties for emergency delay, ordinary delay, queue growth and unnecessary switching/rerouting.
Training setup	Episode-based simulation; fixed seeds for reproducibility; separate validation/test scenarios.
Routing	Recompute a route when edge costs change materially or a route edge becomes unavailable; avoid excessive oscillation.

7. EXPERIMENTAL DESIGN AND EVALUATION

7.1 Scenarios

Scenario	Description	Purpose
S1 — Normal traffic	No emergency; low/medium/high traffic demand.	Check that emergency-aware logic does not unnecessarily harm routine traffic.
S2 — Single emergency	One ambulance or fire engine dispatched through congested intersections.	Measure basic emergency-priority benefit.
S3 — Road blockage	A route edge becomes unavailable during response.	Test dynamic rerouting and response to changing topology.
S4 — Simultaneous emergencies	Two emergency missions with potentially competing routes.	Assess priority handling and route/signal conflicts.
S5 — Disaster disruption	Multiple blocked/high-cost edges or a restricted hazard zone.	Test congestion-aware and hazard-aware route selection.
S6 — Unseen demand/event	Test demand or incident combinations not used in training.	Evaluate generalization and failure modes.

7.2 Baselines

- **B1 — Fixed-time control:** predetermined signal timings.
- **B2 — Actuated/adaptive control:** a conventional traffic-responsive controller where supported by the selected network.
- **B3 — Shortest-path routing:** route by static distance or current edge travel time, without learned coordination.
- **B4 — Emergency-priority heuristic:** rule-based signal priority for an approaching emergency vehicle.
- **B5 — Independent RL:** intersection agents learn without explicit neighbour coordination, if feasible.
- **B6 — Proposed coordinated MARL:** the emergency-aware multi-agent policy.

7.3 Metrics

Metric	Operational definition / interpretation
Emergency travel time	Time from dispatch to incident destination; report mean and distribution by vehicle and scenario.
Emergency response success	Fraction of dispatched emergency vehicles reaching destination within a defined scenario-specific time limit.
Ordinary-vehicle delay	Mean delay per ordinary vehicle compared with free-flow or simulator reference time.
Queue length	Mean/maximum queue length at intersections and over the network.
Network throughput	Vehicles completed or exiting the network per simulation interval.
Route stability	Number of route changes per emergency trip and avoidable route oscillation.
Incident robustness	Performance variation across seeds, demand levels, blocked edges and simultaneous events.
Optional sustainability	Fuel/CO■ estimates only if supported and configured consistently in SUMO.

7.4 Experimental protocol

- Use the same network, demand profiles, emergency dispatches and random seeds for every baseline.

- Use multiple independent simulation seeds per scenario; record the number of runs and report mean, standard deviation or confidence intervals where appropriate.
- Keep test scenarios separate from training and reward tuning.
- Report both absolute metrics and relative change against a named baseline; avoid claiming improvements before the runs are completed.
- Conduct ablations by removing neighbour information or selected reward terms to estimate their contribution.
- Record compute time and training stability, not only traffic outcomes.

7.5 Threats to validity

Simulation may not capture real driver yielding behaviour, communication outages, imperfect incident reports or all safety constraints. Results may depend on the selected road network, demand distribution, reward weights and training seeds. The report will therefore limit conclusions to tested scenarios, publish configuration details where possible and treat real-world deployment as future work.

8. TECHNICAL REQUIREMENTS

8.1 Hardware

Resource	Minimum practical setup	Preferred setup
CPU	Modern 4–6 core laptop/desktop CPU	8+ CPU cores for faster simulation batches
RAM	16 GB	32 GB for larger networks and parallel runs
GPU	Not mandatory for the core SUMO/MARL prototype	NVIDIA CUDA-capable GPU if deep models or vision are added
Storage	10–20 GB free for environments, logs and checkpoints	SSD with additional space for multiple runs and artifacts

8.2 Software and libraries

Tool / library	Purpose	Status
Python 3.10+ (or compatible project environment)	Simulation orchestration, data processing and model code	Core
SUMO + TraCI	Microscopic traffic simulation and programmatic control	Core
PyTorch	Neural policies and learning components	Core if deep RL is selected
RL library (e.g., Stable-Baselines3 or RLlib) / custom MAPPO	Training infrastructure; choose after state/action design	Select one
NumPy, Pandas, Matplotlib / Seaborn	Metrics, result aggregation and plots	Core
NetworkX / OSM-derived network utilities	Graph inspection and routing support if needed	Optional
Git + GitHub	Version control and experiment tracking	Recommended
Streamlit or lightweight web UI	Optional results dashboard and replay controls	Extension
YOLO / XGBoost / LSTM / Transformer / LLM API	Perception, forecasting or GenAI explanation layer	Optional; not required for core claim

8.3 Compute and experiment management

The core research can be conducted on a capable CPU workstation because SUMO is a simulator and small MARL experiments do not inherently require a GPU. A GPU becomes useful if larger neural policies, many parallel training environments or computer-vision models are added. The team should use fixed random seeds, configuration files, model checkpoints, structured CSV/JSON logs and a shared version-control repository. Training, validation and final test runs should be logged separately.

9. EXPECTED OUTCOMES, RISKS AND MITIGATION

9.1 Expected outcomes

- A functioning SUMO-based urban emergency-response testbed with configurable traffic demand, emergency dispatches and road blockages.
- At least three conventional baselines and one learning-based controller, with a documented comparison protocol.
- A MARL controller prototype for coordinated intersection decisions and emergency-aware routing, subject to stable training.
- A results report with emergency travel time, ordinary-vehicle delay, queue length, throughput, route stability and scenario-wise variability.
- A final demonstration and reproducible project artifacts: code, configurations, experiment logs, plots and a technical report.
- A defensible analysis of where coordination helps, where it does not, and what would be needed before real-world deployment.

9.2 Risks and mitigation

Risk	Impact	Mitigation
MARL training unstable or slow	No reliable learned policy	Start with a small grid and limited action space; use a heuristic baseline; test a standard algorithm before custom changes.
SUMO/TraCI integration issues	Delayed experiments	Build and validate a minimal one-intersection scenario early; freeze a known-working environment.
Reward hacking or unfair trade-off	Policy improves one metric while harming others	Report a metric panel; normalize reward terms; run ablations and inspect failure cases.
Scenario complexity too large	Project scope expands beyond timeline	Prioritize ambulance/fire, one network, and single then simultaneous incidents; keep CV/GenAI optional.
Insufficient compute	Few training episodes	Use smaller networks, fewer parallel runs and staged experiments; avoid requiring a GPU for the core prototype.
Weak generalization	Results only work on training scenarios	Hold out demand profiles and incident seeds; report the boundary of generalization.

9.3 Success criteria

The project will be considered successful if the team can reproduce a simulation, run all baselines on matched scenarios, train or implement a clearly specified proposed policy, and report a transparent comparison of emergency and ordinary traffic outcomes. A numerical improvement is a research objective, not a guaranteed outcome. If MARL does not outperform the baselines consistently, a rigorous explanation of the conditions and failure modes remains a valid project result.

10. WORK PLAN AND GANTT CHART

The following schedule is a provisional plan from October 2026 to May 2027. It can be shifted to match the department's actual review dates and project milestones.

Work package	Oct '26	Nov '26	Dec '26	Jan '27	Feb '27	Mar '27	Apr '27	May '27
Literature synthesis and final scope	●	●						
SUMO network + scenario construction	●	●	●					

Work package	Oct '26	Nov '26	Dec '26	Jan '27	Feb '27	Mar '27	Apr '27	May '27
Routing and conventional baselines		●	●					
MARL controller + reward design			●	●	●			
Training, tuning and validation				●	●	●		
Scenario testing + ablation study					●	●	●	
Dashboard / demo integration						●	●	
Analysis, report and final presentation							●	●

10.1 Milestones and deliverables

Target period	Milestone	Deliverable / evidence
Oct–Nov 2026	Scope and literature frozen; SUMO setup validated	Research specification, literature matrix, minimal working scenario
Nov–Dec 2026	Baselines operational	Reproducible fixed/actuated and routing baselines; scenario generator
Dec 2026–Feb 2027	MARL prototype trained and validated	Model checkpoint, reward definition, training curves
Feb–Apr 2027	Evaluation and ablation complete	Matched-seed results, plots, robustness and failure analysis
Apr–May 2027	Final integration and submission	Demo, code/documentation, final report and presentation

10.2 Team work allocation (proposed)

Member	Proposed responsibility
Shubham Singh Negi (23/CS/407)	Research coordination, MARL design/implementation, experiment protocol and technical integration.
Shivansh Dharmshaktu (23/CS/399)	SUMO network/scenario setup, routing baselines, TraCI integration and data logging.
Samar Pratap Singh (23/CS/369)	Metric analysis, plots, ablation/robustness experiments, dashboard/demo and report coordination.
Hardik Gupta (25/A02/063)	Junior team support: literature tracking, scenario testing, documentation and reproducibility checks under team supervision.

These roles are an initial allocation for planning purposes and can be adjusted by the team and guide.

11. REFERENCES

The following five recent works are selected for direct relevance to the project's computer-science core. Four are peer-reviewed journal/conference publications; the wildfire-routing work appears in the ISPRS Archives. Bibliographic details and links should be checked again before final submission, particularly if the department requires a specific citation style.

[1] H. Su, Y. D. Zhong, J. Y. J. Chow, B. Dey, and L. Jin, "EMVLight: A Multi-agent Reinforcement Learning Framework for an Emergency Vehicle Decentralized Routing and Traffic Signal Control System," *Transportation Research Part C: Emerging Technologies*, vol. 142, article 103955, 2022/2023. DOI: 10.1016/j.trc.2022.103955. Focus: joint emergency routing and signal pre-emption.

[2] C. M. Thompson, H. Gao, and A. H. Behzadan, "A reinforcement learning-based routing algorithm for large street networks," *International Journal of Geographical Information Science*, vol. 38, no. 2, pp. 183–215, 2024. DOI: 10.1080/13658816.2023.2279975. Focus: dynamic, safety-aware routing on large networks.

[3] H. Goel, Y. Zhang, M. Damani, and G. Sartoretti, "SocialLight: Distributed Cooperation Learning towards Network-Wide Traffic Signal Control," in *Proceedings of the 22nd International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 2023. Preprint: arXiv:2305.16145. Focus: scalable distributed cooperation among traffic-signal agents.

[4] B. Raei, R. Safarzadeh, and X. Wang, "Congestion-aware Multi-agent Reinforcement Learning for Wildfire Evacuation Routing," *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, vol. XLIX-M-1-2026, pp. 41–50, 2026. DOI: 10.5194/isprs-archives-XLIX-M-1-2026-41-2026. Focus: congestion-aware multi-agent disaster evacuation routing.

[5] W. Jiang, X. Huang, G. Wang, et al., "Multi-agent meta-reinforcement learning based on contrastive predictive coding—For rapid adaptive traffic control after emergencies," *PLoS ONE*, vol. 21, no. 9, e0357779, 2026. DOI: 10.1371/journal.pone.0357779. Focus: state representation and rapid policy adaptation for post-emergency traffic control.

Additional source supplied for background: R. Donatus, K. Ter, and D. Udekwe, "Multi-Agent Reinforcement Learning in Intelligent Transportation Systems: A Comprehensive Survey," arXiv:2508.20315, 2025 (revised 2026). This is a preprint/survey and is not counted among the five core references above. It is useful for taxonomy and limitations of MARL in intelligent transportation systems.

SOURCE AND SCOPE NOTES

1. Project identity, intended capabilities, scenarios, baselines and metrics are derived from the supplied CrisisFlow concept PDF. 2. Submission structure is aligned with the DTU CSE notice dated 29 September 2026. 3. The timeline, team role allocation, hardware recommendations, state/action design and reward formulation are proposed planning choices and should be reviewed with the project guide. 4. No performance gains are claimed for CrisisFlow before experiments are run.

END OF RESEARCH PLAN